Regression models

Module 3 - Lesson 1

Overview

Regression predicts a continuous number from features. Linear regression is the interpretable baseline; regularization and non-linear models extend it.

Key Concepts

- Linear regression fits y = w0 + w1*x1 + ... + wn*xn by minimizing squared error.
- Coefficients measure the expected change in the target per unit of a feature.
- Mean squared error (MSE) and R-squared evaluate fit; compare against a baseline.
- Regularization (Ridge/Lasso) shrinks coefficients and prevents overfitting.
- One-hot encode categorical features; scale numeric features for some algorithms.
- Watch for correlated features (multicollinearity) which destabilize coefficients.

Example

```
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
model = LinearRegression().fit(X_train, y_train)
y_pred = model.predict(X_test)
print(r2_score(y_test, y_pred), mean_squared_error(y_test, y_pred))
```

Summary

Regression predicts continuous outcomes with interpretable coefficients. Evaluate with MSE and R-squared, regularize to avoid overfitting, and always test on held-out data.

Practice Checklist

- Train a linear regression and interpret one coefficient.
- Compare plain linear vs a regularized model on held-out data.
- Plot predictions vs actuals and discuss the spread.